

Our Topic/Scenario: Electoral Analysis, Across Time

Our chosen topic is electoral analysis, focusing on the 21st century. We leveraged standard (structured, tabular) data from the 193 countries recognised by the United Nations (excluding Vatican City) to explore trends, patterns, and dynamics in voter turnout, voting methods, and democracy trends (using PolityV) within global presidential and parliamentary institutions. This approach allowed us to analyse data across diverse political contexts, reaching our aim of addressing critical queries about political participation, and democratic processes. Furthermore, gaining data on a major parliamentary country provided an avenue to explore the accuracy of prediction polling, and using a sample of 52 countries, explored re-election trends. In the light of contemporary challenges such as declining voter turnout, the influence of misinformation, and the evolving nature of electoral systems, our data research may provide a critique of the political state of the world.

After initially choosing the primary attribute of our chosen database to be ISO3, a unique three-letter code which uniquely identifies each country, and ensuring its prevalence among all the data we have collected, we later realised that if the country has multiple rows of data, there would be repetition of these unique primary keys, so we generated an identifying primary key to be unique for the year, month, and day for which elections took place (and in multi-day elections, took the last day of the period). Alongside the date, we added an abbreviation of whether the country is parliamentary (PA) or presidential (PR), leading to primary keys like the following “ALB20130623PA”. We identified our key entities to be Election Participation, Voting Methods, Polity Score, Election Victory, and country-specific data for analysing polls; UK Election Results and UK Election Polls. Some key metrics are Voter Turnout, Voting Percentage, and Date for time-series analysis.

Generalisation: We have generalised the election victory entity that has the sub-entity UK Election Results, and used inheritance principles with opinion polls, resulting in the sub-entity UK Opinion Polls.

Use Cases: We are able to predict election outcomes, by comparing opinion poll winners with election victory winners, and therefore identify patterns in how well polls predict election results to utilise it in future years. Another use case is being able to analyse democracy trends, using the PolityV score to check whether a country’s democracy level impacts the election outcomes, and this can also be predicted in the years to come.

Data

(a) Data Sources

To fulfil the goals of our analysis, we required a large amount of international, time-series data.

Query 1: Voter Turnout

- <https://www.idea.int/data-tools/data/voter-turnout-database>

Query 2: Voting Method

- <https://www.idea.int/data-tools/tools/special-voting-arrangements/data-explorer>

Query 3: Opinion Polls vs. Election Outcomes in the UK

- Election Outcomes: Wikipedia Web-Scraping
- Opinion Polls: <https://www.ipsos.com/en-uk/voting-intentions-great-britain-1997-2002> and <https://www.ipsos.com/en-uk/voting-intention-great-britain-recent-trends>

Query 4: Re-Election

- Wikipedia Web-Scraping

Query 5: Democracy Levels

- PolityV: <https://www.systemicpeace.org/polityproject.html>

(b) Data Selection and Considerations

In determining our analysis of global electoral procedures and analysis, we recognise the 193 Member States of the UN, but exclude the Vatican City due to a lack of data. Furthermore, electoral practices have a long history, and due to the time-restraint of this project, coupled with our desire to tandem explore sentiment data, our analysis is one of the 21st Century.

Have you identified publicly available data sources? Are they reliable and representative of a real scenario?

We have identified International IDEA ("an intergovernmental organization (IGO) with a mandate to support sustainable democracy worldwide"), which holds many databases, including the "Voter Turnout Database" described as "the best resource for a wide array of statistics on voter turnout from around the world". It is a comprehensive global collection of voter turnout statistics from presidential and parliamentary elections since 1945, also including European Parliament elections.

Will you generate synthetic data (including as a complement to your real datasets)? Have you identified suitable tools for that (for instance, Chat-GPT or other AI tool) and do know how to prompt them to get useful outputs?

We will not use Chat-GPT or other AI tools to generate synthetic data, however we have used AI tools to edit or format our dataset for better efficiency

Is there sufficient information that allows you to understand how the data was captured and is structured? Here, you may account for any bias or missing data that can influence and/or restrict the type of queries you can run and/or the results.

In this case, specifically regarding our aim to analyse changes in voter participation within and across countries over time, we have recognised over 500 missing values within the International IDEA data. These present themselves either as NaN or - . This is particularly worrying for our analysis when these values occur within the Voter Turnout, Total Vote, Registration, VAP Turnout, and Voting Age Population columns. Due to difficulties with imputation, due to the multiple confounding factors that influence not only democratic processes, but data collection, we have decided to keep these missing values in our data, doing our best to avoid any repercussions within our analyses.

Any restrictions or specific concerns about the chosen data sources?

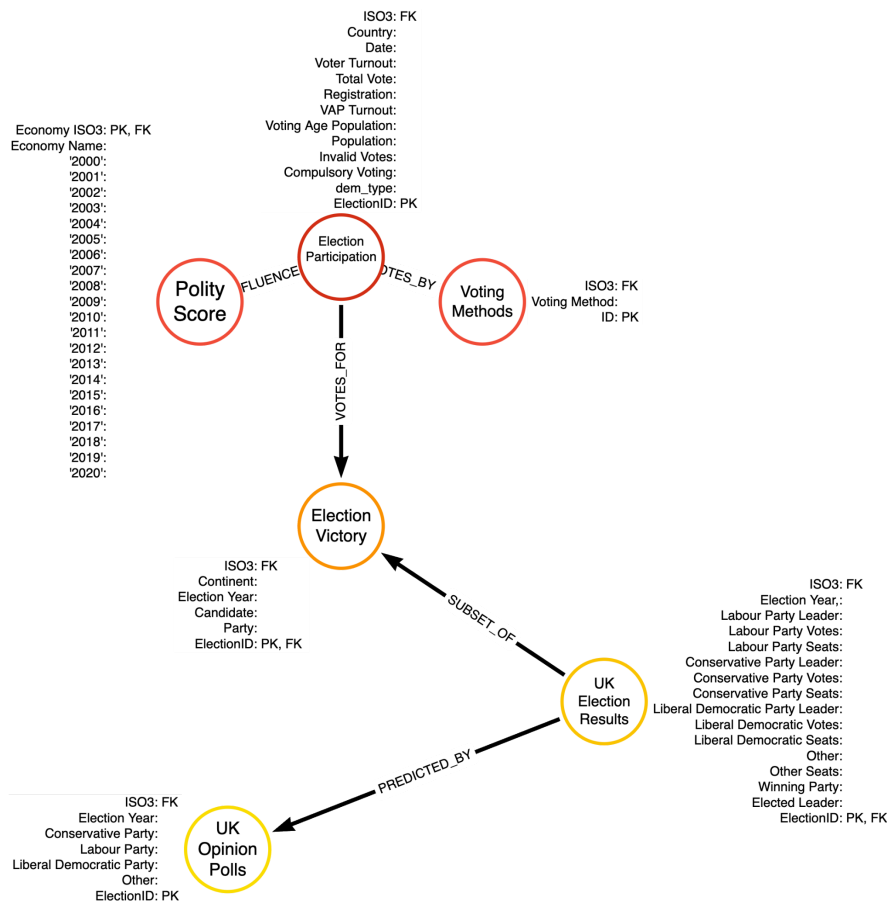
Our concerns about the chosen data sources stem from these missing values (discussed previously)

(c) Data Collection

We followed a manual process while collecting the data from our data sources in part (a), in order to organize our data for each query, and have easily readable CSV files that we later on used in our relational models. For query 1, we started off by collecting the data from IDEA's Voter Turnout Database, and the data collection was quite simple as we just had to download the readily available dataset and then divide the dataset into presidential and parliamentary systems. This made it simpler to turn the data into organized CSV files. For query 2, a similar process was followed as query 1, but a difference was that the available data was in the form of interactive maps rather than a downloadable dataset. Therefore, we collected data on each individual SVA (Special Voting Arrangement) and asked ChatGpt to condense all the data and organize it into a single CSV file which contained all the data. For opinion polls in query 3, most of our data was from Wikipedia but we also had data from Ipsos, where we did web scraping and made individual CSV files on all data, later merging them into single, organized CSV files. We followed a more tedious process of data collection in query 4, as we had over 50 sample countries in which we had to manually scrape data for. We first set a seed and wrote a code to select a randomised stratified sample. We used Wikipedia as a source for all of the countries, and collected data on each election year's party and candidate

winners. For specific countries in the sample like San Marino, there was no available yearly election data, so we randomised once more to get a country to substitute, from the same continent. This helped us generalize our results for the entire population as our sample was representative. For query 5, we utilised the Polity website and downloaded the dataset that was provided.

Database Modelling



The image describes all the entities and relationships from our database used for electoral analysis.

The entities in our model are:

- (1) *Election Participation*: Contains information on voter turnout by country (UN member states), over time.
- (2) *Voting Methods*: Contains the voting method for each country (UN member states).
- (3) *Polity Score*: Contains the Polity2 measure for each country over the period 2000 - 2020.
- (4) *Election Victory*: Contains the winning party/leader in each country chosen from our stratified sample.
- (5) *UK Election Results*: Contains UK election results, i.e. the year, the party leader for each running party, vote shares, etc.
- (6) *UK Opinion Polls*: Contains opinion poll data classified into 'Labour', 'Conservative', and 'Other'.

The relationships in our model are described as follows:

- (1) *Election participation* is connected to *voting methods* because it connects the voters in election participation to the voting method they used in the election

- (2) *Election participation* is connected to *Polity score* because we believe that turnout (electoral participation) is influenced by the country's regime type, measured by polity score on the scale -10 → 10 (authoritarian → democratic)
- (3) *Election participation* is connected to *Election victory* because it connects voter turnout to *who* the voters voted for, and who won, both over time.
- (4) Both *UK election results* are connected to *Election victory*, as they are both subsets of *Election victory*, recording which party won each year, however for each of the two countries separately and in more detail.
- (5) *UK opinion polls* connects to *UK election results*, because we believe the opinion poll helps to predict the election result.

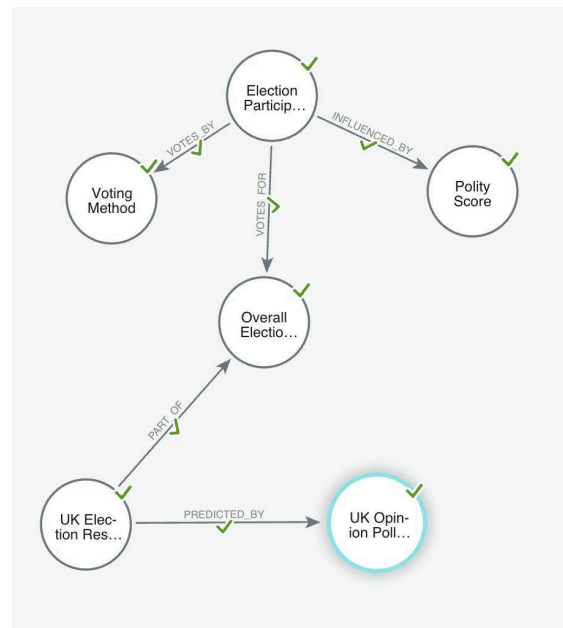
Database Creation and Design

Interface used:

We used The **Neo4j** interface in order to *create and design* our database. Instead of using the cypher queries to populate our database, we converted all of our files to CSV files and imported them into the database, after creating the nodes and relationships as seen above in the screenshot, modelled after our data modelling using the *arrows app*.

Issues faced:

- (1) We were making ISO3 codes the primary keys for all nodes, however there were multiple entries for each country, and therefore the primary key was not unique to each data entry
- (2) After using the interface to build the relationships, not all of them were showing up in our actual database, which was affecting our queries and not letting us match nodes.

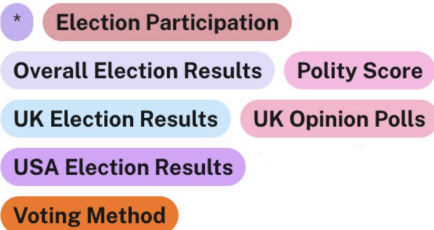


Solutions undertaken:

- (1) We utilized CHATGPT to make an 'ElectionID' column in the entity 'Election Participation', which would uniquely identify each data entry based on the ISO3 code, the year of the election, the day of the election, and PA for parliamentary elections, and PR for presidential elections. We then input these unique ElectionID's in all the other entities, ensuring unique primary keys.
- (2) We manually created the relationships using cypher queries, as depicted below:

Database information

Nodes (2,473)



```
MATCH (e:`Election Participation`), (v:`Voting Method`)
WHERE e.ISO3 = v.ISO3
CREATE (e)-[:VOTES_BY]->(v);
```

Relationships (6,283)



```
MATCH (ukr:`UK Election Results`), (oer:`Overall
Election Results`)
WHERE ukr.ISO3 = oer.ISO3
CREATE (ukr)-[:PART_OF]->(oer);
```

Property keys

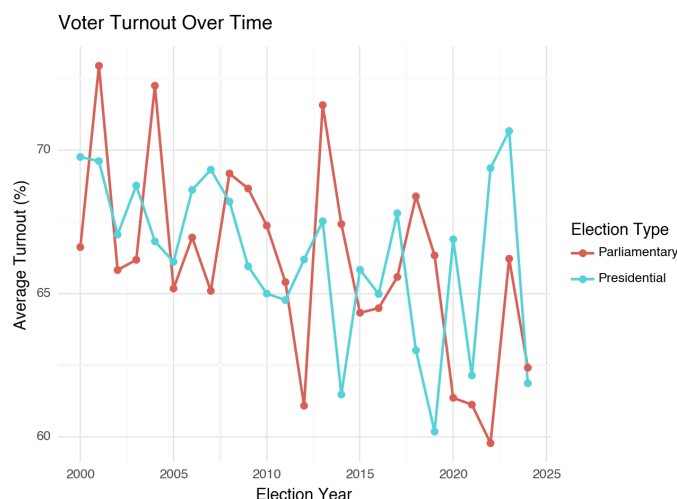


```
MATCH (ukr:`UK Election Results`), (oko:`UK Opinion
Polls`)
WHERE ukr.ElectionID = oko.ElectionID
CREATE (ukr)-[:PREDICTED_BY]->(oko);
```

We successfully created the correct amount of nodes and relationships, in line with our data.

Query Descriptions and Analysis of Results (available at [Queries.ipynb](#))

Query 1: How Does Voter Participation Vary Among Institution Type, Region, and Across Time?



Our objective with this query is to analyse changes in voter participation within and across countries over time. We analysed whether a type of institution (parliamentary or presidential) influences voter turnout, and then we analysed this over time, in order to see if voter turnout increased/decreased more in a specific democracy type.

Input parameters: The data includes the previously stated key attributes, and we also filtered the data by election year to analyse trends over time and by ISO3 (country).

Additionally, countries were characterised by electoral systems and compulsory voting laws, in order to make a more accurate comparison.

Output analysis: from both the graph and the textual output with samples: We can summarise from the graph and textual output that both parliamentary and presidential countries had fluctuations considering the average voter participation, but we can observe that presidential election voter turnout has faced a larger decline than parliamentary, in the past 5 years, whereas parliamentary election turnout depicts an increasing trend until approximately 2024.

Query 2: Does the Number of Special Voting Arrangements (SVA's) Available influence Voter Turnout?

It is well documented that in-person voting is available across all countries, but special voting arrangements (SVAs) allow voters to exercise their right to vote by alternative means (i.e. instead of casting a ballot in-person, on election day or across the election period, in the default polling station). We classified SVA's into 3 categories, (1) Proxy Voting, (2) Remote Voting, (3) Postal Voting, (4) Mobile Voting, and (5) Remote Electronic Voting. We counted the number of voting methods per country through the 'Voting Method' entity, (barring the countries with compulsory voting because it is a confounding variable affecting our causal question), and then analysed the voter turnout from the 'Election Participation' entity. We organised our output to show the aggregate voter turnout for countries which offered 1 SVA, 2 SVA's, and so on, to infer how the number of SVA's influences voter turnout. We predicted that voter turnout and the number of SVA's would have a **positive correlation**.

Input parameters:

The data needed includes attributes that are related to Voting Methods (mobile, postal, proxy, etc.) and corresponding Voter Turnout rates. We have filtered the data by voting method in order to compare the participation rates from each country with their respective voting methods. The condition for filtering will involve selecting countries where these SVAs were actively used at least once in a recent election (after 2000).

Output analysis: Below is our output

Number of SVA's	Average Voter Turnout (2 d.p)
1	65.68%
2	63.28%
3	56.35%

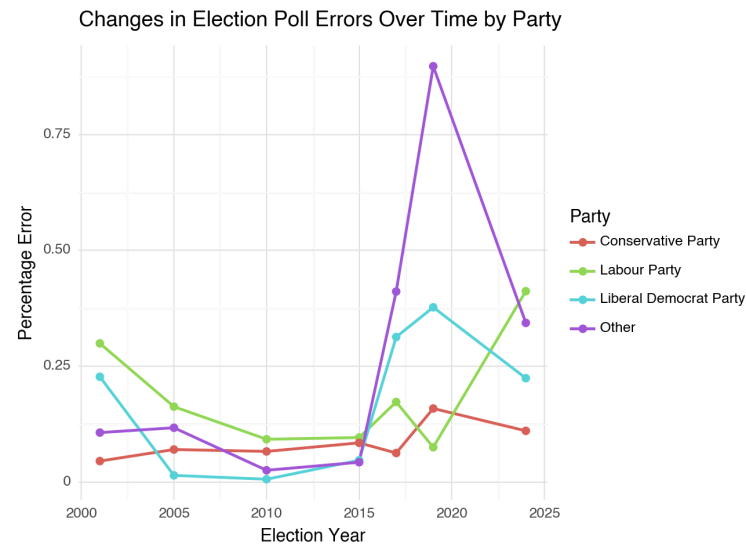
According to this output, there is a **negative correlation** between the number of SVA's and Average Voter Turnout, contrary to what we predicted. This indicates that as the number of Available special voting methods *increases*, the average voter turnout *decreases*.

Query 3: How Well Do Opinion Polls Predict Election Outcome?

In this query, our objective was to choose a case study to decide how well opinion polls predicted election outcomes. The reason we chose the UK as our case study was because we found the most available, easily accessible data that contains multiple years' information on elections. We also believed that the opinion polls

in the UK would be one of the most accurate as there are advanced methods of data collection and much more reliable, and therefore would produce more accurate outputs. Input parameters: The data for this query includes opinion poll data (from different sources we have specified) and election results for the corresponding years, both for the United Kingdom. The condition for filtering includes the polling data's accuracy and the margin of error to ensure that we have compared accurate polls to actual results.

Output analysis:



We can summarise from the output of this query that there have been significantly higher election poll errors for smaller parties (voting for candidates that are neither labour nor conservative party are on the rise) in the UK. We can also say that 0.25% error is not large enough to make a prediction and therefore opinion polls for larger parties are good predictors as they are all below that values, whereas predictors for other parties are less accurate.

Query 4: What are the countries with the highest re-election rates in both Parliamentary and Parliamentary Elections?

Our objective with this query is to analyse which are the most stable parliamentary and parliamentary democracies in the world, through a sample of countries. Our measure for stability is the rate at which the same Candidate or Party has been re-elected between the years 2000 - 2024. When selecting which countries we should analyse, we wanted the selected countries to be as representative of the general population as possible, and so we selected a stratified sample of countries in each continent proportional to the total number of countries in that continent.

Input parameters:

The input data includes election results across multiple years for countries that we have randomly generated for our sample with our code. We filtered the dataset to only show the year, candidate, and party who have won that year, so we were able to see if there are any candidates and/or parties that have been in parliament/the president for multiple consecutive terms. We also used data from the entity 'Election Participation' to retrieve the country name by matching ISO3 codes

Output analysis:

Country_Name	Party_or_Candidate	Election_Type	ReElection_Count
Israel	Likud	Parliamentary	8
Palau	Independent	Parliamentary	7
Switzerland	Swiss People's	Parliamentary	6

Luxembourg	CSV	Parliamentary	5
Greece	New Democracy	Parliamentary	5
Belarus	Alexander Lukashenko	Presidential	5
Azerbaijan	Ilham Aliyev	Presidential	5
Bulgaria	Georgi Parvanov	Presidential	4
Bulgaria	Rumen Radev	Presidential	4
Congo	Denis Sassou Nguesso	Presidential	4

This output shows us the 5 most ‘stable’ parliamentary and presidential democracies. However, on further inspection of the outputted countries, we recognise that the fact that just because a party or candidate has been in power for many years, does not indicate a ‘stable’ democracy, but it may actually indicate anti-democratic ruling of the country, and other confounding variables such as election manipulation, intimidation tactics, and voting method could be influencing the party or candidate staying in power.

Query 5: How Have Democracy Levels Evolved Over Time and Across Political Regimes?

Our objective in this query is to analyse whether there were changes in democracy levels of countries and if so, what may be the reasons behind that. We analysed each country’s regime change percentage from the Polity Scores, and also their regime type. For other queries in the future, we would be able to analyse whether the change in regime has affected other factors, like voter turnout, or new data that we want to correlate this to. **Input parameters:** the data for this query includes the Polity Scores for each country over the course of multiple years. The dataset was filtered to a time period (2000-2020) to be more homogeneous with other query datasets. As the polity score already organizes countries by regime type, and where on the spectrum from democratic to authoritarian each country is, therefore we have been able to get data on regime type changes over the years as well. We coded the query so that we could observe the highest and lowest percent changes in democracy for countries, and their regime type.

Output analysis:

We can observe from the output that the most democratization in countries is usually seen in democratic regimes, with the exception of Comoros, which is an authoritarian regime, but still has democratized at a very high rate. And as expected, the countries with the highest scores of negative percent change are mostly authoritarian, again with an exception that is the democracy Iran.

Country Code	Percentage Change	RegimeType
ZMB	500.0	Democratic
GHA	300.0	Democratic
GEO	250.0	Democratic
COM	200.0	Authoritarian
DJI	200.0	Democratic
GIN	-500.0	Authoritarian

TZA	-400.0	Authoritarian
KGZ	-366.66666666666663	Authoritarian
HTI	-350.0	Authoritarian
IRN	-333.33333333333337	Democratic

Database Technology

Choice of Database Software/Tool

For this project, we decided to work with Neo4j, a graph database, as it most fit the nature of the relationships within our data. Unlike relational databases, which are efficient with structured data, graph databases like Neo4j are ideal for handling complex relationships between entities - which is an aspect that is key in our analysis of electoral data. For example, Election Participation is connected to Voting Methods, Polity Scores, and Election Victories. These relationships can be directly modeled as nodes and edges in Neo4j, making it easier to perform complex queries about the connections between countries and voting systems.

To Create the Graph Model

First, we created nodes such as Election Participation, and other entities, alongside relationships between them. Then, we imported our CSV files and added them manually to the entities we created. We were then able to create relationships between entities to different variables in CSV files which were connected in entities.

Reproducibility Instructions

1. Install Neo4j
 - a. Downloading from the official Neo4j website or,
 - b. You can use Neo4j Aura, which is a cloud-based version of the software
2. Create the graph database
 - a. First we open Neo4j desktop and create a new database
 - b. Then we are able to run the cypher queries to create the nodes and relationships for our entities
3. Load data
 - a. We are able to store our CSV files in the import directory of Neo4j, then run the LOAD CSV command to import the loaded data into the database
4. Run queries
 - a. After we have loaded our data, we were able to use cypher queries to analyse relationships between countries, voting methods, and other variables
5. Exporting the graph
 - a. Exporting the graph can be done as a GraphML file, which is what we utilised

Technical Issues

Neo4j is usually optimised for relationship-based queries, which may cause performance issues with large datasets. But, for our project, as our data was organized and carefully cleaned to a manageable size, this was not an issue.

A Statement on the Use of Generative AI Tools, As Per School and Course-Specific Policies

- (1) Merging the presidential and parliamentary into one file:
<https://chatgpt.com/share/679765bb-65d0-8002-8357-dab5be3372c3>
- (2) Making a unique 'ElectionID' for each row in the entity 'Election Participation' based on the ISO3 code, the year of the election, the day of the election, and PA for parliamentary elections, and PR for presidential elections.